

CSC3044S
Artificial Intelligence:
Agent Architectures

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Agent Architectures

Agent Architectures – Definition I

- “[A] particular methodology for building [agents]. It specifies how . . . the agent can be decomposed into the construction of a set of component modules and how these modules should be made to interact. The total set of modules and their interactions has to provide an answer to the question of how the sensor data and the current internal state of the agent determine the actions . . . and future internal state of the agent. An architecture encompasses techniques and algorithms that support this methodology.”

  - Pattie Maes (1991)

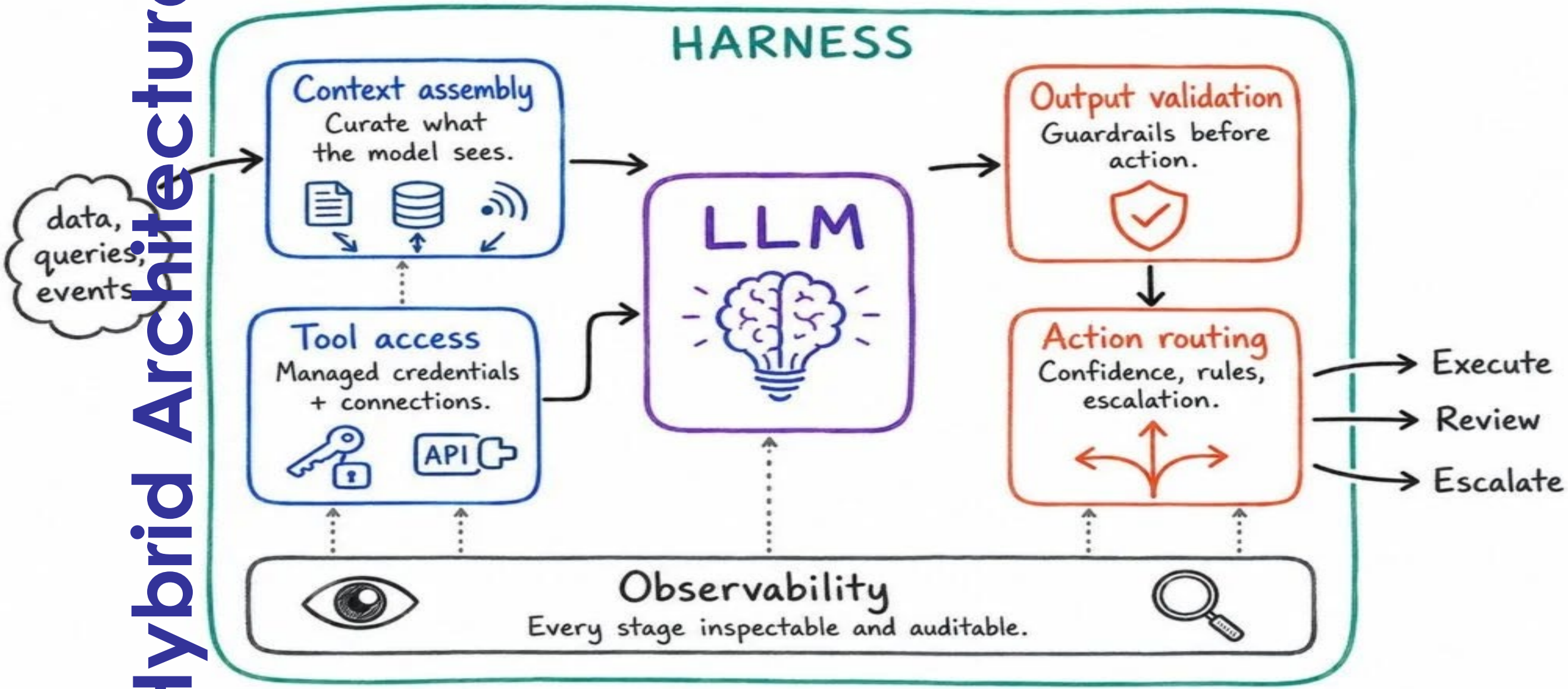
Agent Architectures – Definition II

- "'[A] specific collection of software (or hardware) modules, typically designated by boxes with arrows indicating the data and control flow among the modules. A more abstract view of an architecture is as a general methodology for designing particular modular decompositions for particular tasks."

- Leslie Kaelbling (1991)

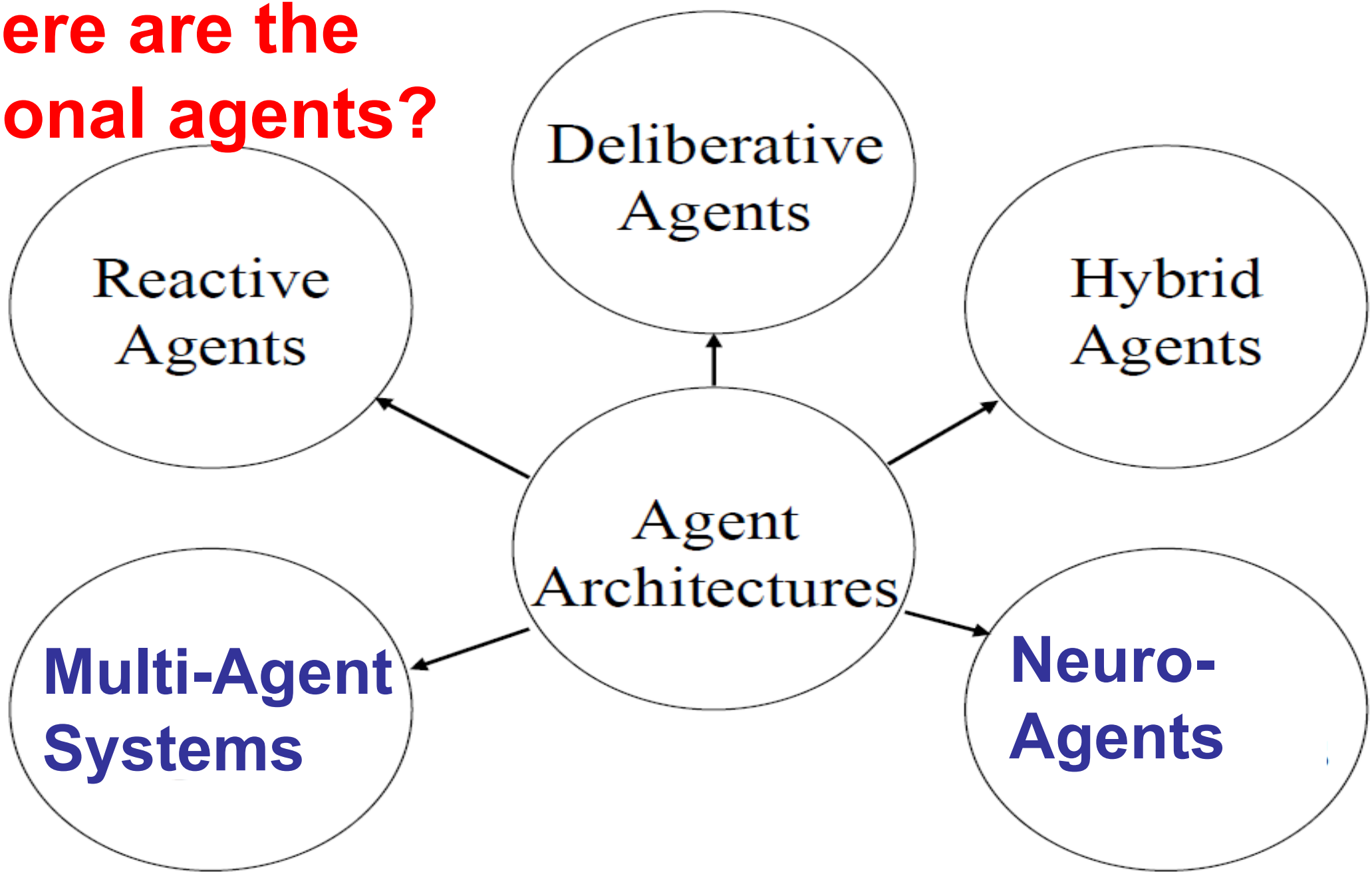
Agent = Model + Harness

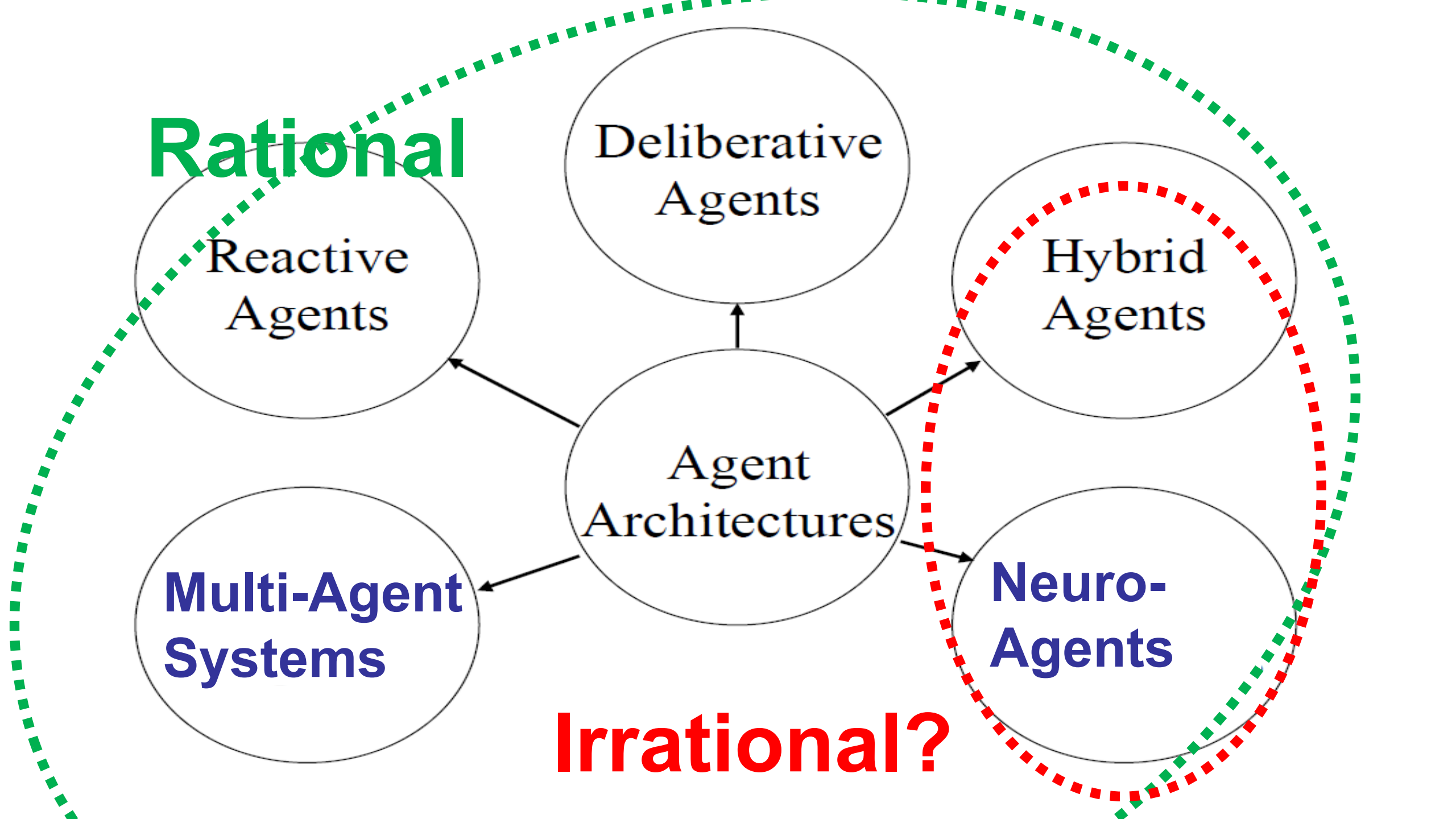
Hybrid Architecture



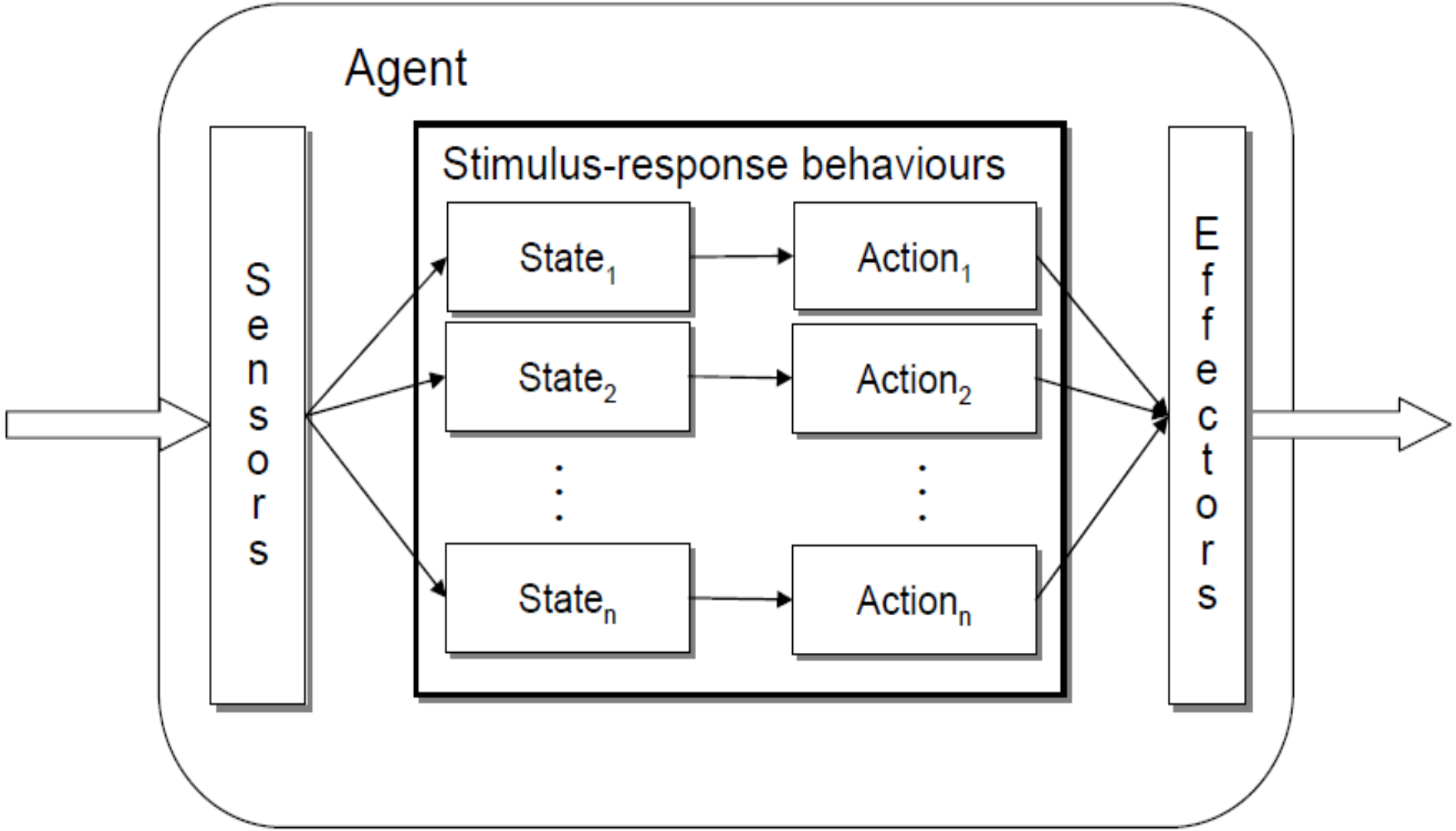
The model reasons. The harness governs everything else.

Where are the rational agents?




Rational
Deliberative Agents
Reactive Agents
Hybrid Agents
Agent Architectures
Multi-Agent Systems
Neuro-Agents
Irrational?

Recall – Reactive Agents



Reactive Agent Architectures

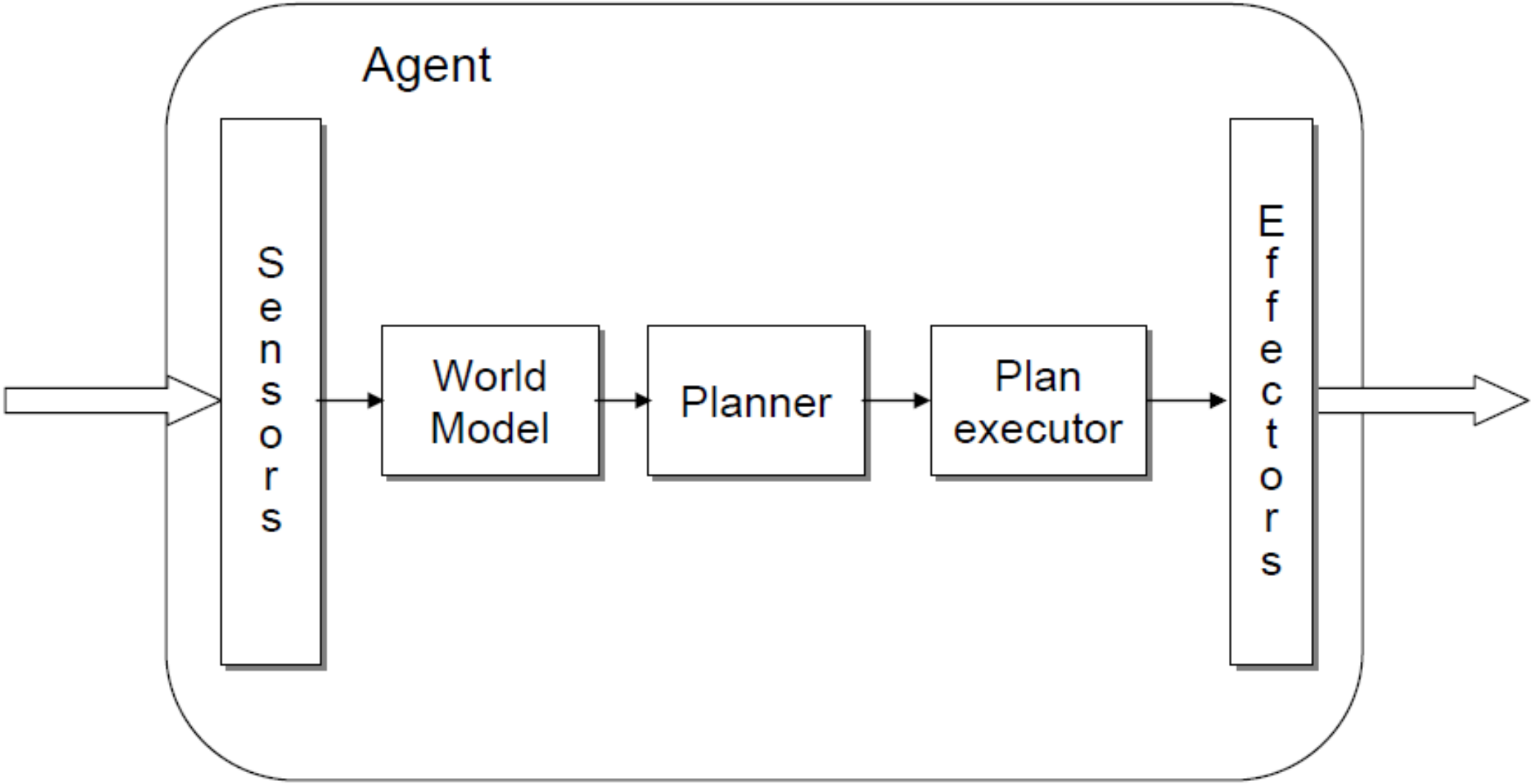
- Reactive agents have at most a very simple internal representation of the world – provide tight coupling of perception and action.
- Behaviour-based paradigm.
- Intelligence is a product of interaction between an agent and its environment.

Reactive Architectures – Examples

- Subsumption architecture[1].
- Some (emergent behaviour) multi-agent system models:
  - Boids[2]
  - Stigmergy model[3]
  - Few others!

1. Brooks, R. (1991) "Intelligence Without Representation", Artificial Intelligence 47: 139-159.
2. Reynolds, C. (1987). "Flocks, herds and schools: A distributed behavioral model". Proceedings of the 14th annual conference on Computer graphics and interactive techniques. ACM. pp. 25–34.
3. Theraulaz, G., & Bonabeau, E. (1995). Coordination in Distributed Building. Science 269: 686-688.

Recall – Deliberative Agents



Deliberative Agent Architectures

- **Framework for Sense-plan-act:** How agents sense their environment, generate a plan, and act on plan.

- Deliberative agent behaviour dictated by:

  - ***Environment state***, ***internal state***, and ***plan*** (logical reasoning, symbolic manipulation, rule-set, ANN, …).

  - ***Maps inputs → outputs***.

Deliberative Agent Architectures – Examples

- Belief, Desire, Intention (BDI) Architecture[1].
- Shoham: Agent-Oriented Programming[2].
- Deductive reasoning agents[3].

1. Rao, A., and Georgeff, M. (1991). Modeling Rational Agents within a BDI-Architecture. In Proceedings of the 2nd International Conference on Principles of Knowledge Representation and Reasoning, pages 473–484.
2. Shoham, Y. (1990). Agent-Oriented Programming (Technical Report). Stanford University: Computer Science Department.
3. Walton, C. (2006), Practical reasoning and deductive agents, Agency and the Semantic Web. https://doi.org/10.1093/oso/9780199292486.003.0010

Deliberative Agent Example: Deductive Reasoning

- How can an agent decide what to do? (use theorem proving).

- Basic idea is to use logic to encode a theory stating the best action to perform in any given situation – Let:
  - ***p*** be this theory (Set of rules);
  - **Δ** be a logical database describing the current state of the world;
  - ***Ac*** be the set of actions the agent can perform;
  - $\vdash_p \Phi$ means that Φ can be proved from using **Δ**.

Deductive Reasoning Agents

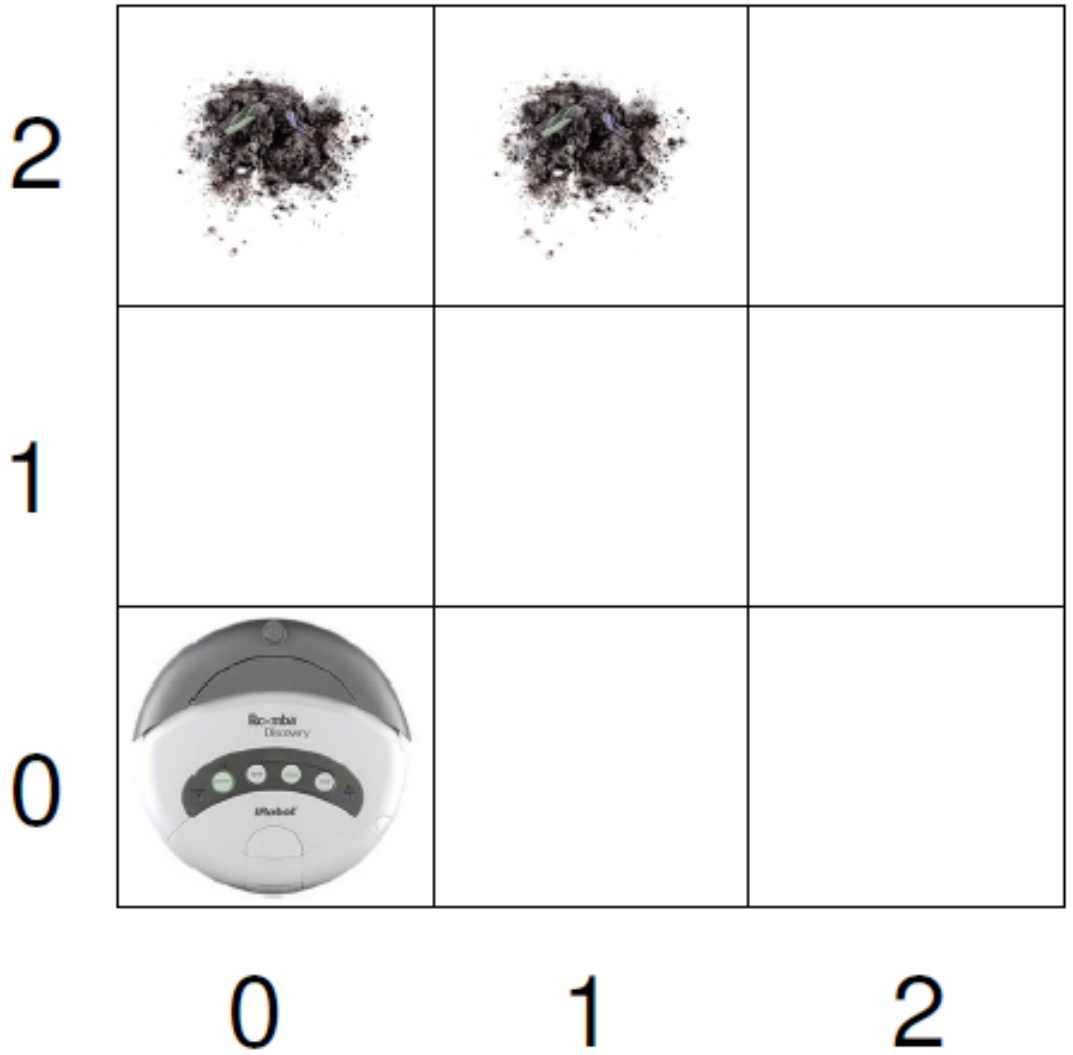
for each $\alpha \in Ac$ do /* *try to find an action explicitly prescribed* */
    if $\Delta \vdash_\rho Do(\alpha)$ then
        return α
    end-if
end-for

for each $\alpha \in Ac$ do /* *try to find an action not excluded* */
    if $\Delta \nvdash_\rho \neg Do(\alpha)$ then
        return α
    end-if
end-for

return `null` /* *no action found* */

The Vacuum World

- Goal is for the robot to clear up all dirt.

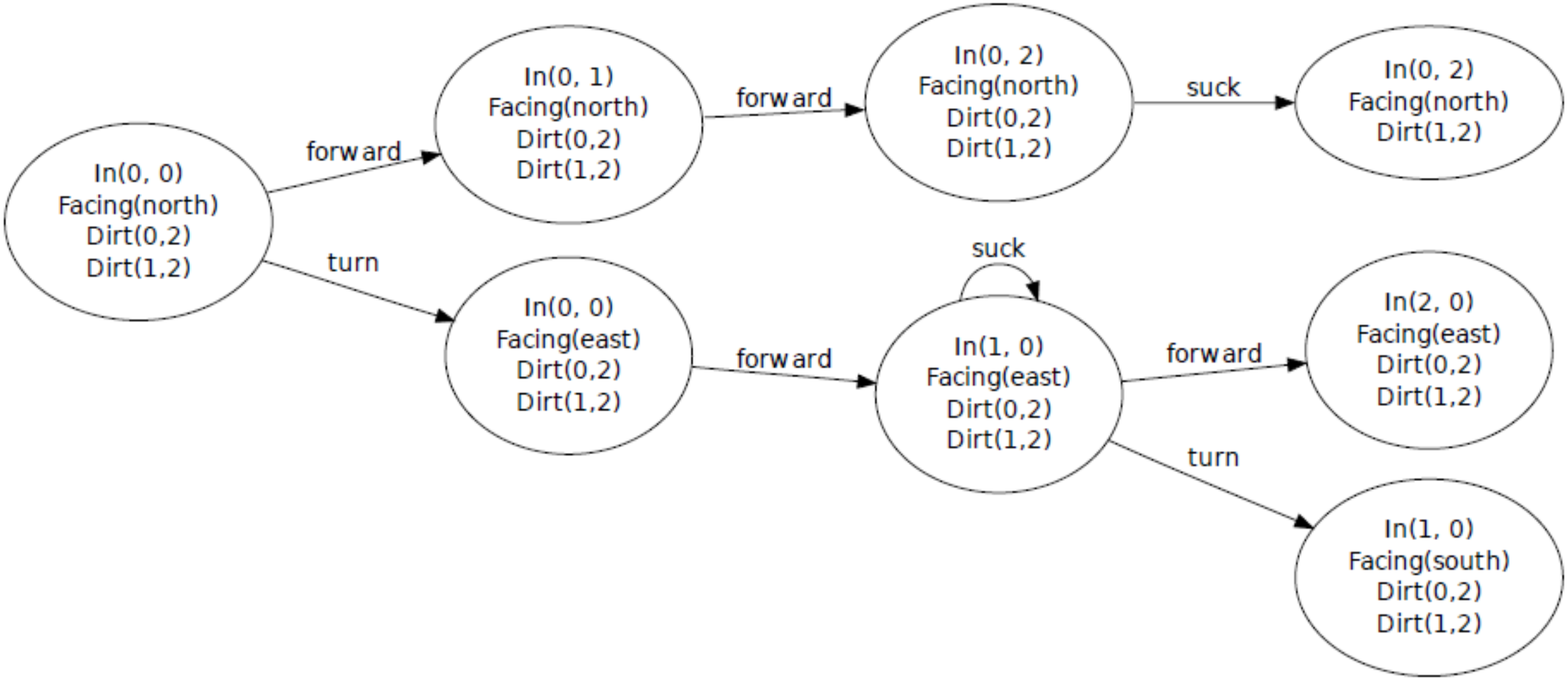


The Vacuum World

- Use three (3) domain predicates in this task:
  - ***In(x, y)*** agent is at *(x, y)*;
  - ***Dirt(x, y)*** there is dirt at *(x, y)*;
  - ***Facing(d)*** the agent is facing direction *d*.

- Possible actions:
  - ***Ac*** = *{ turn, forward, suck }*.
  - “Turn” means turn right.

The Vacuum World – Example Run



The Vacuum World

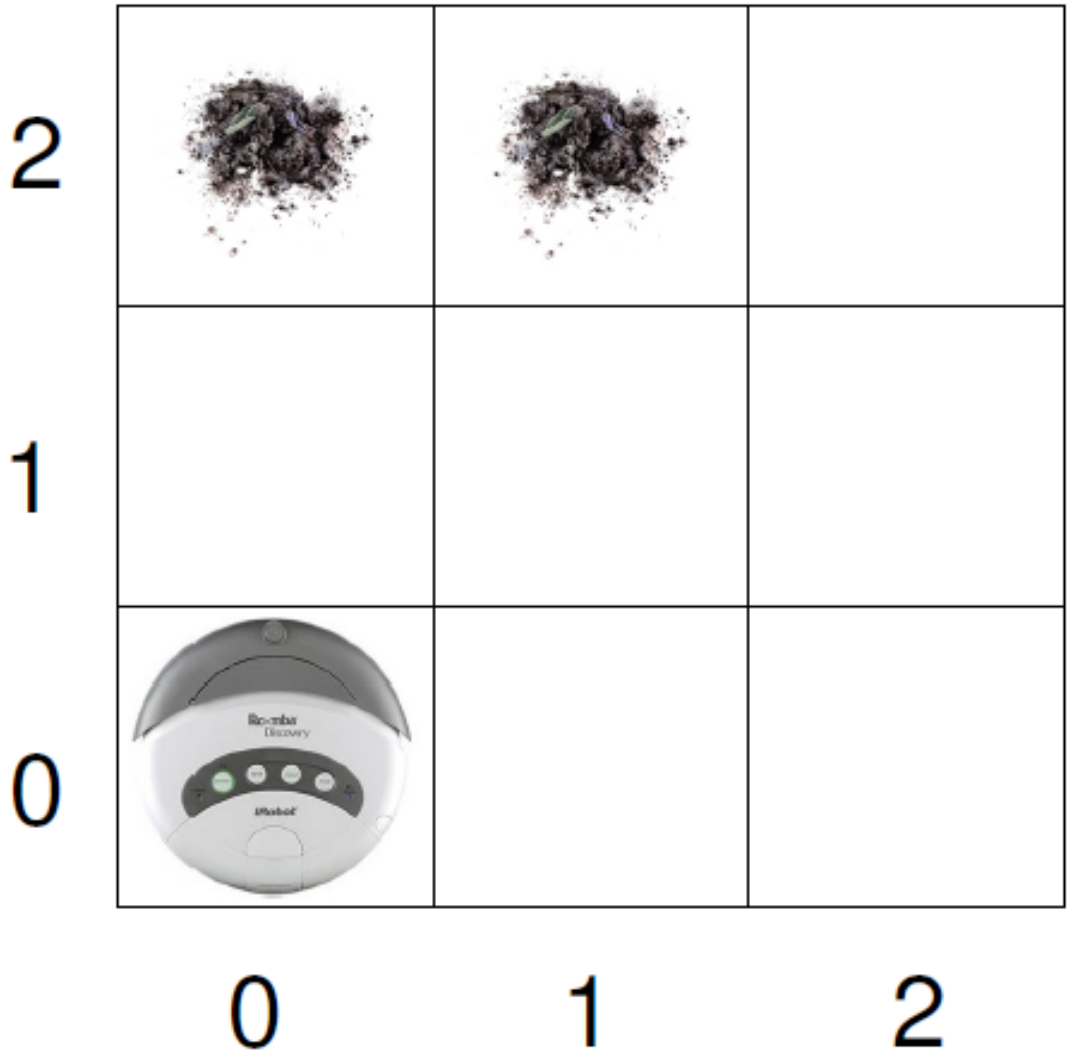
- Rules ***p*** for determining what to do:

$$In(0,0) \wedge Facing(north) \wedge \neg Dirt(0,0) \longrightarrow Do(forward)$$
$$In(0,1) \wedge Facing(north) \wedge \neg Dirt(0,1) \longrightarrow Do(forward)$$
$$In(0,2) \wedge Facing(north) \wedge \neg Dirt(0,2) \longrightarrow Do(turn)$$
$$In(0,2) \wedge Facing(east) \longrightarrow Do(forward)$$

- IF robot at In(0, 2) THEN reset its position to (0, 0)
- Starting at (0, 0) the robot will clean up dirt – Problems? – Is there a better way to do this?

The Vacuum World

- Goal is for the robot to clear up all dirt.

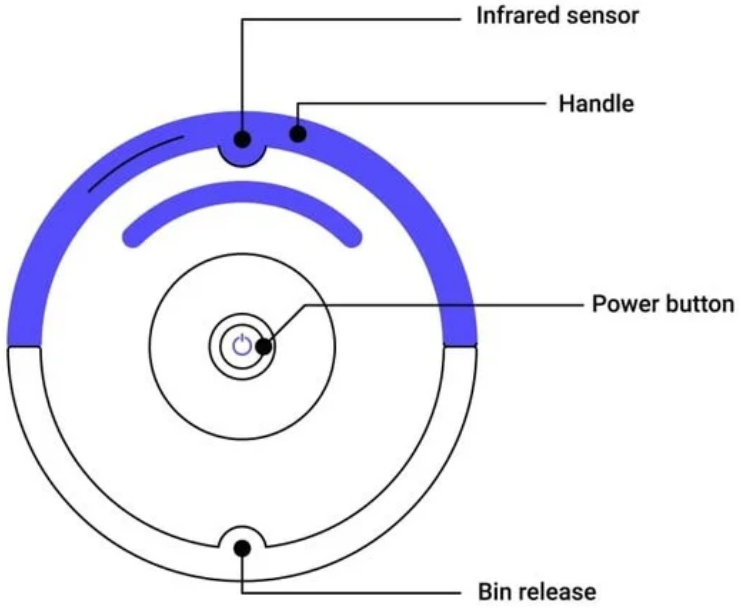


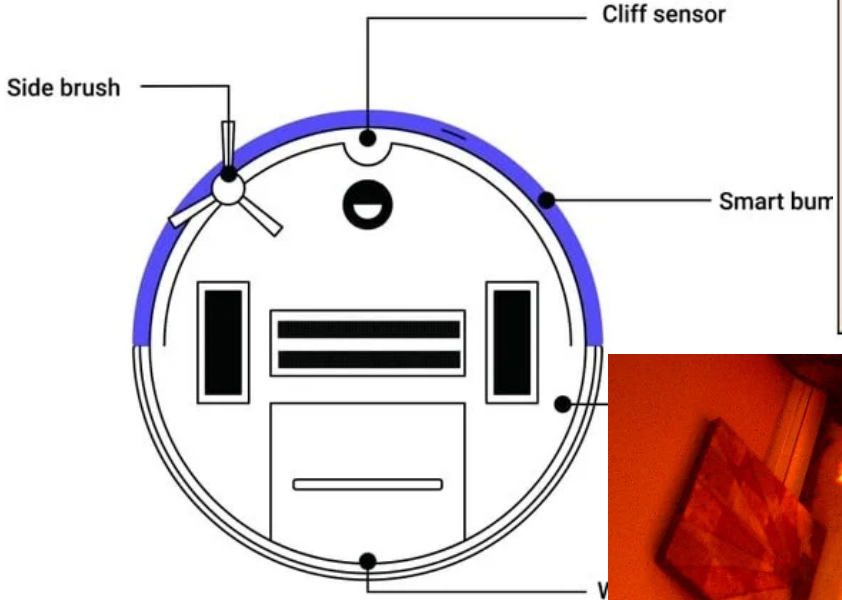
The Vacuum World – Problems?

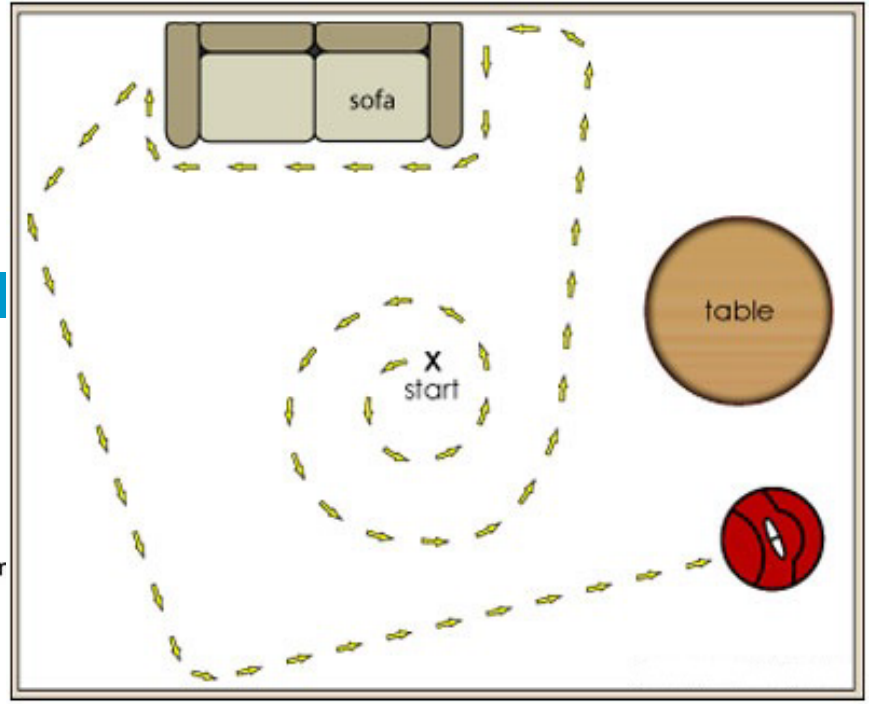
- How to convert video camera input to Dirt(0, 1)?
- Decision making assumes a static environment.
- Decision making with *first-order logic*, e.g., ∀x (is-carpet(x) → is-floor(x) ; is-floor(x) → cleanable(x)); ..., is impractical.
- *Propositional logic*, e.g., T/F statements (e.g., if x=carpet, x=adjacent, x=dirty THEN clean(x); ELSE IF…) decision making is, worst case, NP-complete:
  - e.g., Like the *Boolean Satisfiability Problem* – Time to find a solution increases exponentially with *n*.

Vacuum World: Better Approach?

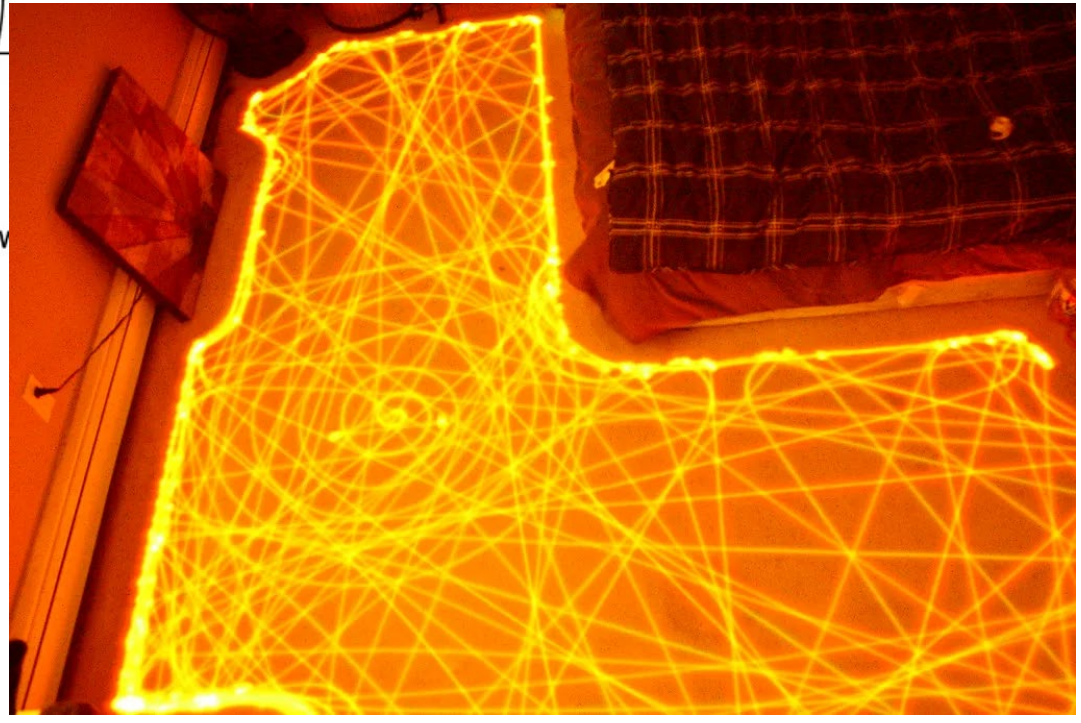
Vacuum World: Roomba







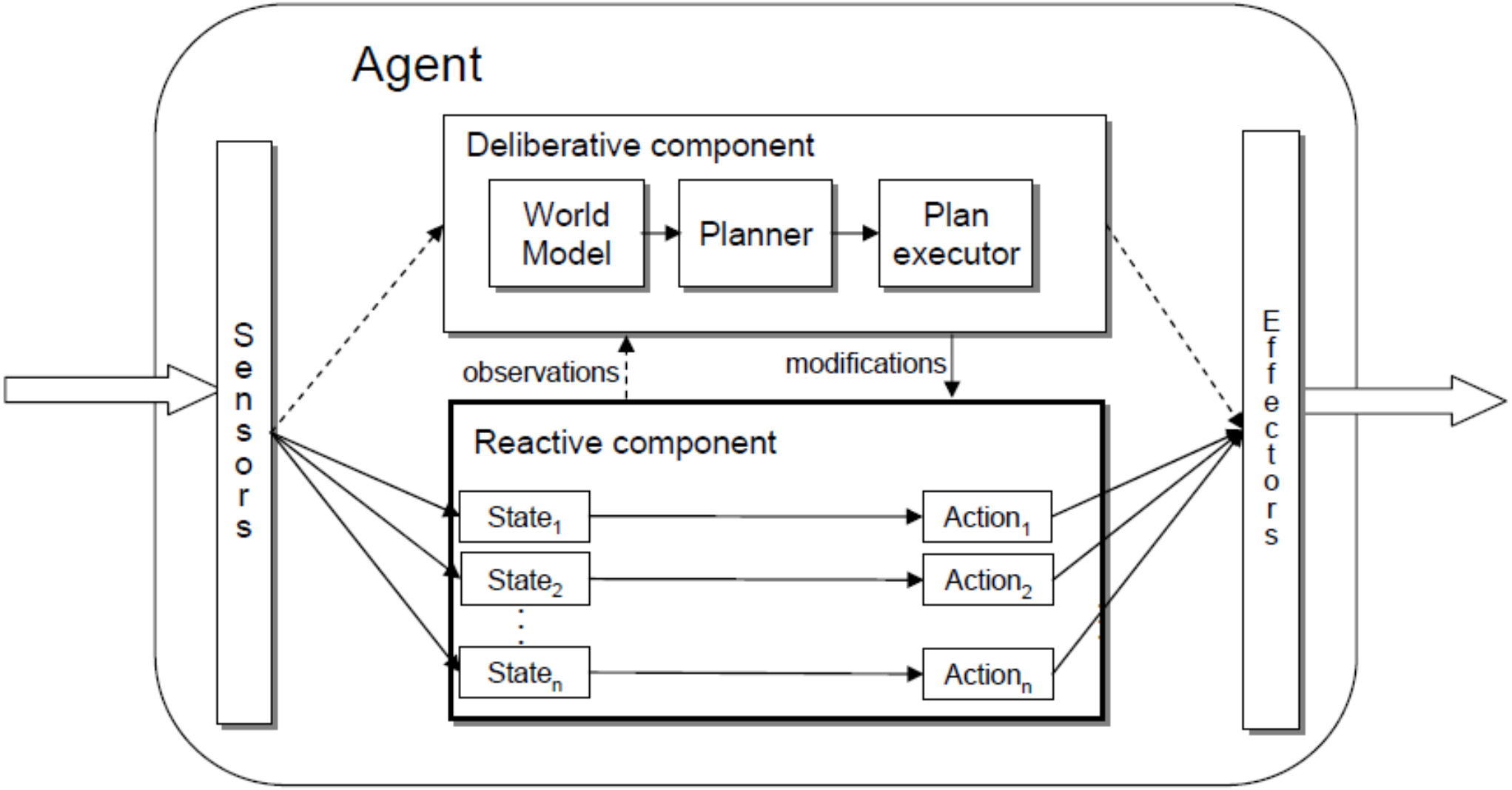




Hybrid Agent Architectures

- An agent consists of:
  - **Deliberative component:** Subsystems make decisions using cognitive (e.g., neuro-based), or symbolic (e.g., logic-based) reasoning.
  - **Reactive component:** Subsystems that react quickly to events without complex reasoning.

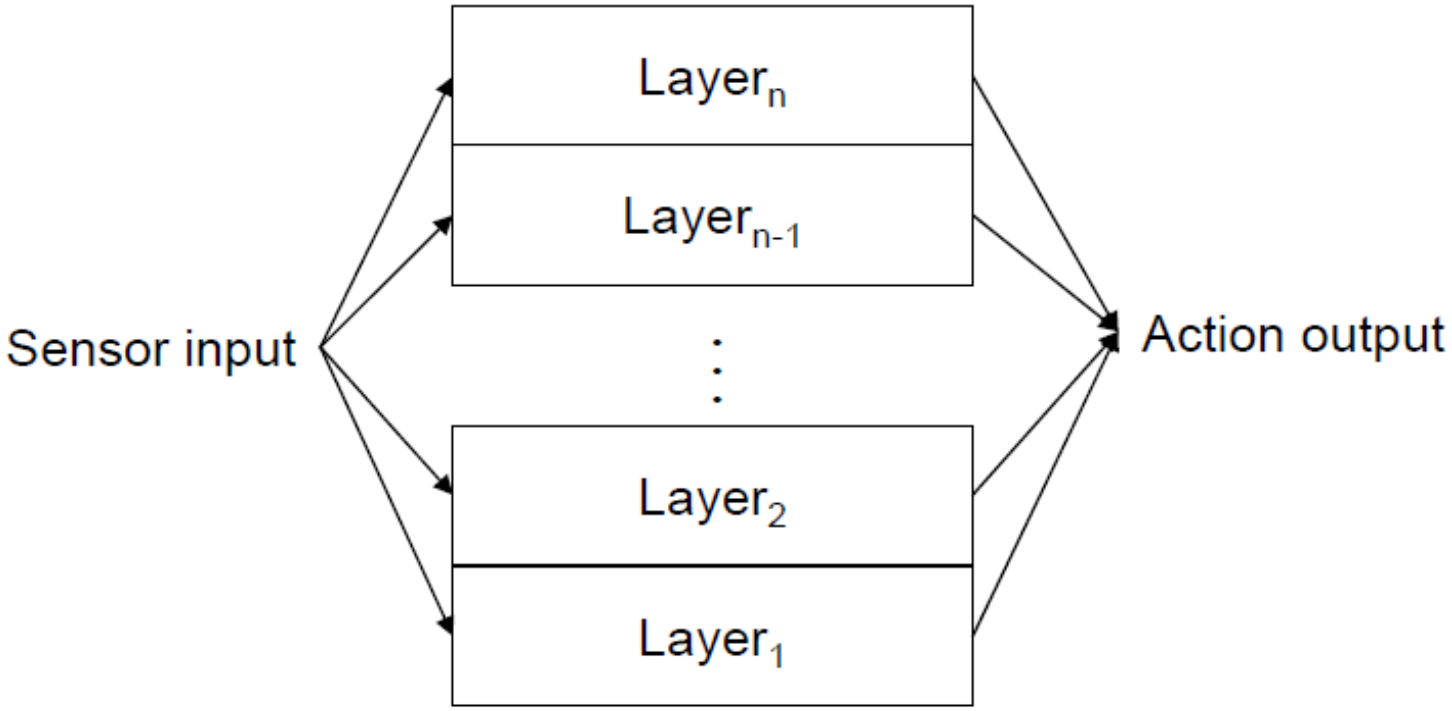
Hybrid Agent Architecture

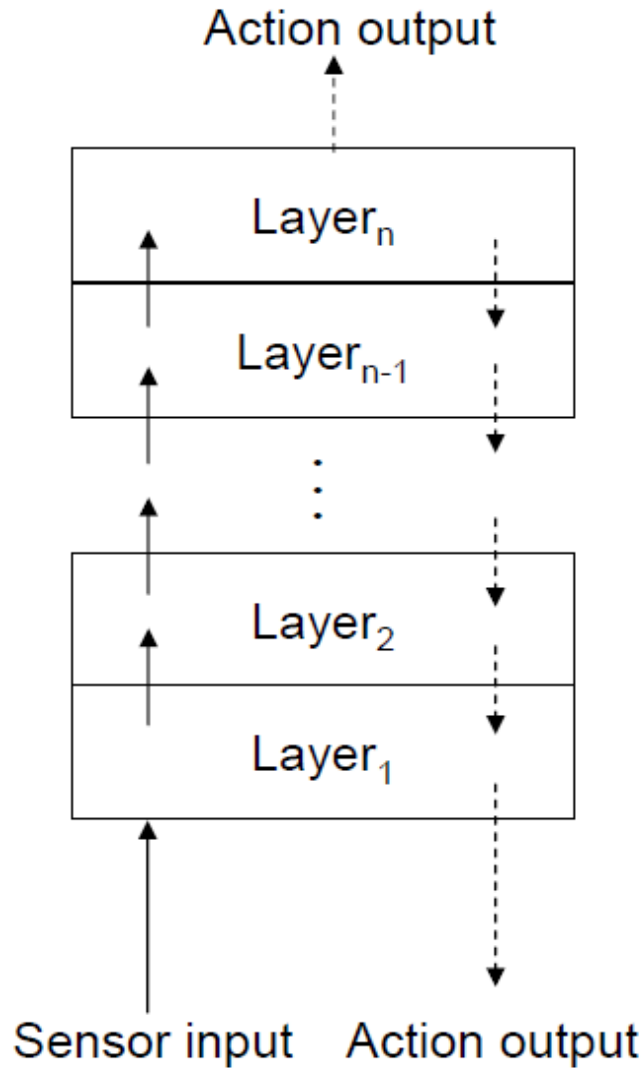


Hybrid Agent Architectures

- Combine **reactive** and **deliberative** components and form a hierarchy of interacting layers.
- Each layer reasons at a different level of abstraction.
- Two types of layering:
  - Horizontal layering
  - Vertical layering.

Layered Hybrid Agent Architecture



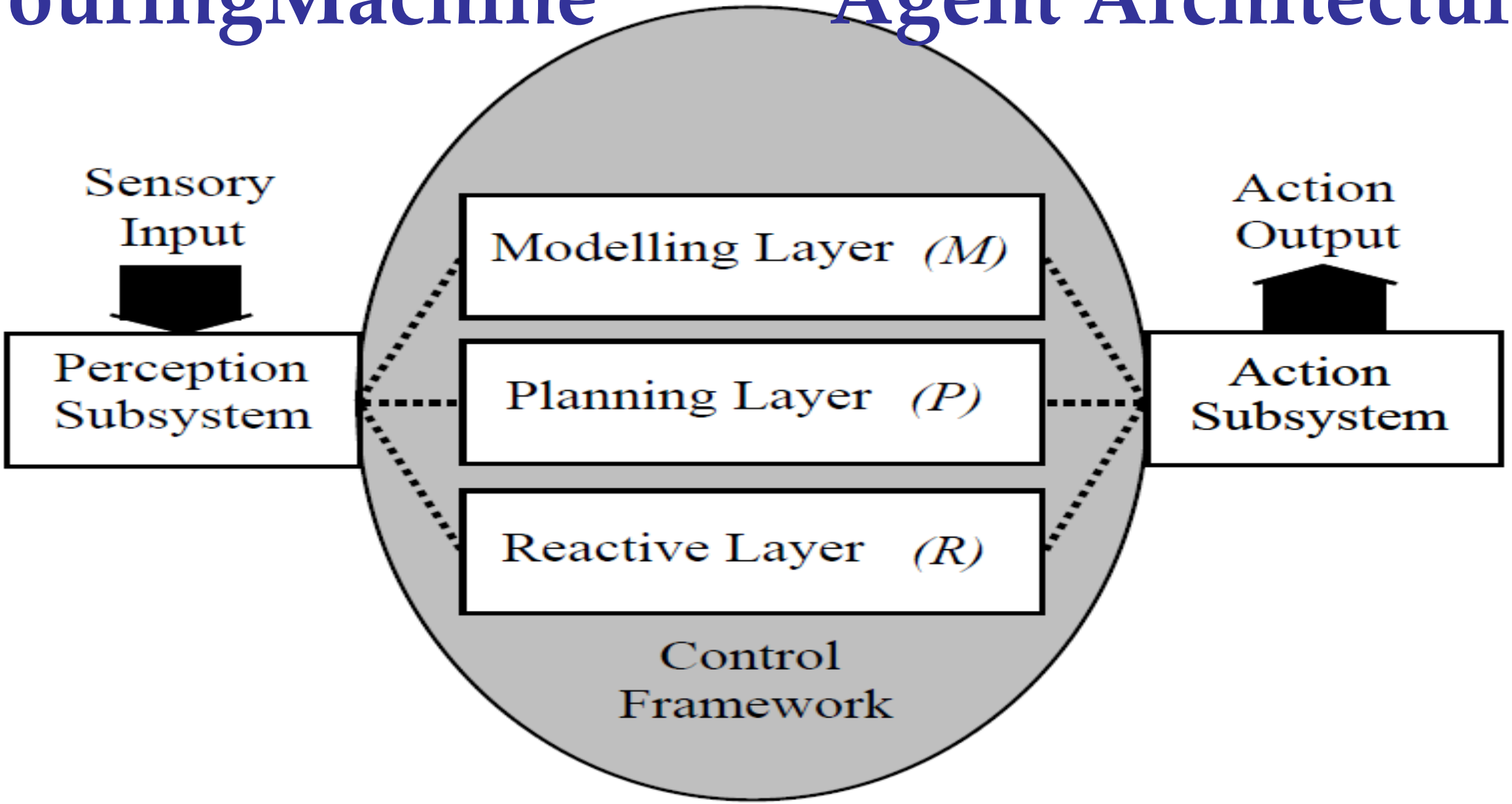


Example: TouringMachines[1]

- Hybrid structure makes it suitable for complex, dynamic environments – e.g.:
  - **Autonomous Robotics:** Robots require the Reactive layer for obstacle avoidance and the Planning layer for navigation.
  - **Multi-Agent Systems:** Effective for traffic control simulations, automated factory floors, or distributed problem-solving nodes.

1. Ferguson, I. (1992). TouringMachines: An architecture for dynamic, rational, mobile agents, PhD Thesis. Computer Laboratory, University of Cambridge, Cambridge, UK.

TouringMachine Agent Architecture



Planning Layer: Robot behaviour:
Transport Package to drop-point (x, y)




Planning Layer: Robot behaviour:
Transport Package to drop-point (x, y)
Modeling Layer: Map of warehouse floor


Planning Layer: Robot behaviour –
Transport Package to drop-point (x, y)
Modeling Layer: Map of warehouse floor
Reactive Layer: Avoid collisions


CGTN

Layered Architecture – Problems?

Layered Architecture – Problems?

- How do we decide what layers are active and when?

Layered Architecture – Problems?

- Mediator function (central control system):
  - If there are *n* layers – each capable of *m* possible actions – there are *mn* interactions.
- But – The mediator function can introduce a bottleneck into the agent's decision making.

Robot Warehouse Packing Sorting

- Active Layers: Planning, Modeling, Reactive? – Mediator function:

WHILE package NOT dropped // ***Planning layer active***

WHILE collision NOT imminent // ***Reactive layer inactive***

Map path (next package) pick-up → drop-off; // ***Modeling layer active; Planning layer inactive***

Go; // ***Planning layer active; Modelling layer inactive;***

DO

DO

RESTART

Reactive Architectures

- Many unsolved problems associated with symbolic AI – This led researchers to question the viability of the whole paradigm – thus developing ***reactive*** (and then ***hybrid***) architectures.

- Current (popular) agents (e.g., Claude, Gemini, GPT…) are all ***deliberative (sense-plan-act) architectures*** – but are overly complex for many real-world problems, and unsuitable for others.

- Simplest approach is purely ***reactive*** (*sense-react*) architecture.

Intelligence needs Embodiment?

- Two perspectives on how to achieve general intelligence:
  - Computational functionalism.
  - Embodied cognition.
- Functionalists believe intelligence is complex information processing independent of substrates and referents (Newell & Simon, 1976).
- Recent advances in (LLM) agents support this hypothesis based on LLMs' emergent reasoning abilities (Wei et al., 2022).

Embodied Cognition

- Proponents of *embodied cognition* argue intelligent agents must interact with the real world – embodiment is necessary to give meaning to symbols.
- This is the "**Symbol Grounding Problem**" (Harnad, 1990).
- From this viewpoint – intelligence requires real-world (physical) interaction (Brooks, 1991).

Embodied Cognition II

- “What separates humans from machines is the fact that we exist in situations, not apart from them, and that we do so in an embodied and non-conceptual way.” (Dreyfus, 2006).

- Again – This is the Symbol Grounding Problem.

(Current) Agent Text Generation

- Lack of symbol grounding is not so evident.
- LLMs can get away with vague, general statements – the reader's imagination fills in the gaps.

please explain a practice described by the following sentence:
"engineers working on a runway, aided by drones"

The practice described in the sentence refers to the use of drones in supporting engineers while they perform maintenance, inspections, or construction tasks on a runway. Here's how it typically works:

1. **Inspection and Monitoring**: Drones are equipped with cameras and sensors to capture high-resolution images, videos, and data. Engineers can use these to inspect the runway for cracks, wear, or any other structural issues without needing to physically walk the entire length of the runway. This saves time and provides a comprehensive overview of the condition of the surface.
2. **Surveying**: Drones can help survey large areas of land, like runways, quickly and efficiently. They are often used to capture aerial images for mapping the area, assessing drainage systems, and identifying other potential issues like debris or obstacles.
3. **Data Collection**: Drones can be fitted with various sensors like thermal imaging or LiDAR to detect issues that might not be visible to the naked eye, such as hidden structural problems or temperature changes in certain areas of the runway.
4. **Maintenance and Repairs**: Drones can also assist in more practical ways, such as delivering tools or equipment to engineers in hard-to-reach places, ensuring faster response times when a runway needs urgent attention.

In short, drones enhance the ability of engineers to work more efficiently, safely, and accurately by providing real-time data and reducing the need for human intervention in potentially hazardous areas.

Agent Image Generation

- Image output (DALL-E-3):

- “engineers working on a runway, aided by drones”.



Agent Image Generation

- AI systems do not exist in the world (in an embodied sense).
- They experience things through absorbed coping and inference.
- AI processing has no ontological grounding (real-world experience).



Next:
Subsumption Architecture

What you should Know

- Difference between reactive, deliberative and hybrid agents.
- The subsumption architecture – what it is, its benefits and limits.

Reading (On Amathuba!)

- Russell, S., & Norvig, P. (2021). *Artificial Intelligence: A Modern Approach* – 4th Edition – Ch2 (Intelligent Agents): aima.cs.berkeley.edu/
- Poole, D., & Mackworth, A. (2023). *Artificial Intelligence: Foundations of Computational Agents* – Ch1:https://artint.info/3e/html/ArtInt3e.Ch1.html
- Brooks, R. (1991), Intelligence Without Representation, *Artificial Intelligence*, 47(1991): 139-159.
- Newell, A., & Simon, H. (1976). Computer science as empirical inquiry: Symbols and search. Communications of the ACM, 19(3): 113–126.
- Wei, J., et al. (2022). Emergent abilities of large language models. arXiv:2206.07682.
- Harnad, S. (1990). The symbol grounding problem. Physica D: Nonlinear Phenomena, 42(1-3): 335–346.
- Gahrn-Andersen, R. (2025). Beyond symbol processing: the embodied limits of LLMs and the gap between AI and human cognition. *AI and Society*, 40: 3105–3107.

